

ÖGOR Summer-Workshop for PhD-candidates and Post-Docs

An introduction to Julia and JuMP for Operations Research

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Topic 8

Data structure (part 2)

Tuple, dictionary, set



Constructing and accessing to a tuple

A tuple is an `immutable` array with 1 dimension

Definition:

Let a collection of n items: $item_1, item_2, \dots, item_n$

$(item_1, item_2, \dots, item_n)$

Examples:

```
julia> ("Austria", "France", "Belgium")
```

```
julia> primeNumbers = (2, 3, 5, "seven", 11)
```

Accessing to a tuple:

Same of a vector.

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Accessing to a tuple:

Same of a vector.

Constructing and accessing to a tuple

A tuple is **immutable**

```
julia> vct = [2,7] # array
```

```
julia> push!(vct,3)
```

Right!

```
julia> tpl = (2,7) # tuple
```

```
julia> push!(tpl,3)
```

Wrong!

Constructing and accessing to a dictionary (1/3)

A dictionary is an 1D array where the sequence of elements is **not ordered**

Definition:

Let a collection of n couple *key-value*

```
Dict(key1 => value1, key2 => value2, ...)
```

Example:

```
julia> Dict{43 => "Austria", 32 => "Belgium", 33 => "France"}
```

Constructing and accessing to a dictionary (1/3)

A dictionary is an 1D array where the sequence of elements is **not ordered**

Definition:

Let a collection of n couple *key-value*

```
Dict( $key_1 \Rightarrow value_1$ ,  $key_2 \Rightarrow value_2$ , ...)
```

Example:

```
julia> Dict{43 => "Austria", 32 => "Belgium", 33 => "France"}
```

Constructing and accessing to a dictionary (1/3)

A dictionary is an 1D array where the sequence of elements is **not ordered**

Definition:

Let a collection of n couple *key-value*

```
Dict( $key_1 \Rightarrow value_1$ ,  $key_2 \Rightarrow value_2$ , ...)
```

Example:

```
julia> Dict{Int{<div data-bbox="459 922 536 977" data-label="Page-Footer">The logo for ÖGOR, featuring the word "ÖGOR" in a stylized font with a red swoosh underneath.
```

Constructing and accessing to a dictionary (2/3)

Empty dictionary:

```
Dict()
```

```
julia> code = Dict()
```

Add an entry:

```
nameDictionary[key] = value
```

```
julia> code[49] = "Germany"
```

Delete an entry:

```
pop!(nameDictionary, key)
```

```
julia> pop!(code, 33)
```

Constructing and accessing to a dictionary (3/3)

Return an iterator over all keys in a dictionary:

```
keys(dict)
```

```
julia> keys(code)
```

Return an iterator over all values in a dictionary:

```
values(dict)
```

```
julia> values(code)
```

Constructing and accessing to a set (1/2)

A set is a collection of unordered, unique values

Empty (zero-elements) set:

```
Set()
```

```
julia> a = Set()
```

Initialise a set with values:

```
nameSet([value1, value2, ..., valuen])
```

```
julia> a = Set([1,2,2,3,4])
```

Constructing and accessing to a set (1/2)

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Empty (zero-elements) set:

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Initialise a set with values:

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julia> a = Set([1,2,2,3,4])
```

Constructing and accessing to a set (2/2)

Some operations:

```
union(set1,set2)
```

```
julia> union([1, 2], [3, 4])
```

```
intersect(set1,set2)
```

```
julia> intersect([1, 2, 3], [3, 4, 5])
```

```
setdiff(set1,set2)
```

```
julia> setdiff([1,2,3], [3,4,5])
```

Summary

	mutable	ordered collections
array	yes	yes
tuple	no	yes
dictionary	yes	no
set	no	no

Data structure

Strings



Constructing and accessing to a string (1/4)

A string is an **immutable** finite sequence of characters

```
julia> s1 = "A string."           # ASCII characters
```

```
julia> s2 = """Also a string."""
```

```
julia> """A "special" string."""
```

```
julia> "\u2200 x \u2203 y"           # unicode characters
```

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Constructing and accessing to a string (2/4)

String interpolation

Use the \$ sign

- ▶ to insert existing variables into a string and
- ▶ to evaluate expressions within a string.

```
julia> city = "Linz"
```

```
julia> "Hello $city"
```

```
julia> zc = 4020
```

```
julia> "zc of $city:  $zc, $(zc+10), $(zc+20)"
```

Constructing and accessing to a string (2/4)

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Constructing and accessing to a string (3/4)

String concatenation

Two manners:

- ▶ use the `string()` function:

```
julia> string("Hello ", city)
```

`string()` converts non-string inputs to strings

- ▶ use the `*` operator:

```
julia> "Hello " * city
```

Constructing and accessing to a string (4/4)

```
julia> s1 = "A string."
```

```
julia> s1[3]
```

```
julia> s1[3:5]
```

```
julia> s1[begin:3]
```

```
julia> s1[3:end]
```

```
julia> s1[3] = 'b' # error!
```

```
julia> s1 = s1[begin:3] * "w" * s1[5:end]
```

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